

Sistemas de Numeración,

Operaciones lógicas,
y otros...

Fundamentos de
Arquitectura de Computadoras

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DECBINASCII
06501000001A
06601000010B
06701000011C
... ..
09701100001a
```

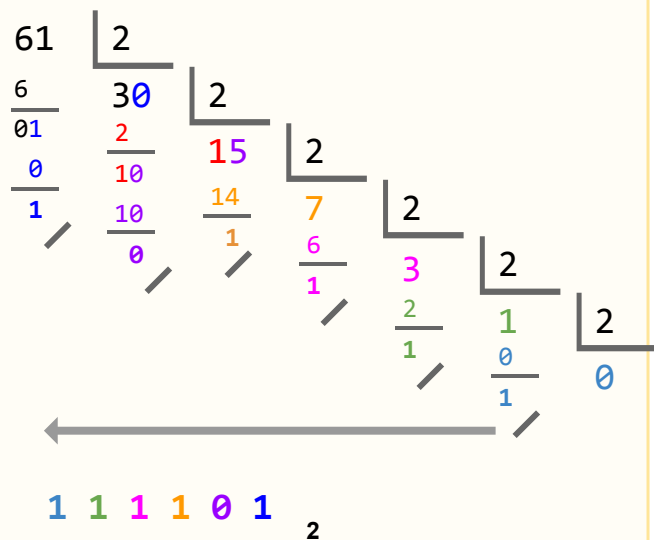
Sistemas de Numeración

Decimal	Binario	Octal	Hexadecimal
0	0000 0000	00	00
1	0000 0001	01	01
2	0000 0010	02	02
3	0000 0011	03	03
4	0000 0100	04	04
5	0000 0101	05	05
6	0000 0110	06	06
7	0000 0111	07	07
8	0000 1000	10	08
9	0000 1001	11	09
10	0000 1010	12	0A
11	0000 1011	13	0B
12	0000 1100	14	0C
13	0000 1101	15	0D
14	0000 1110	16	0E
15	0000 1111	17	0F
16	0001 0001	20	10
17	0001 0010	21	11
18	0001 0011	22	12
19	0001 0100	23	13

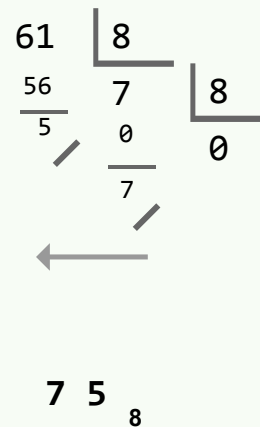
Conversión decimal a...

Ejemplo 61_{10}

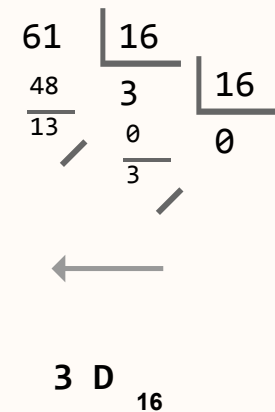
→ **Binario (Base 2)**



→ **Octal (Base 8)**



→ **Hexadecimal (Base 16)**



Conversión a Decimal

Binario $\rightarrow$ Decimal

$$\sum_{i=0}^n d_i \cdot 2^i$$

Ejemplo: 111101_2

$$1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 =$$

$$1 \times 32 + 1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 61$$

Conversión a Decimal

Binario $\rightarrow$ Decimal

$$\sum_{i=0}^n d_i \cdot 2^i$$

Ejemplo: 111101_2

$$1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 =$$

$$1 \times 32 + 1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 61$$

ejemplos simples:

$$101_2 =$$

Conversión a Decimal

Binario $\rightarrow$ Decimal

$$\sum_{i=0}^n d_i \cdot 2^i$$

Ejemplo: 111101_2

$$1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 =$$

$$1 \times 32 + 1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 61$$

ejemplos simples:

$$101_2 = 5_{10}$$

Conversión a Decimal

Binario $\rightarrow$ Decimal

$$\sum_{i=0}^n d_i \cdot 2^i$$

Ejemplo: 111101_2

$$1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 =$$

$$1 \times 32 + 1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 61$$

ejemplos simples:

$$101_2 = 5_{10}$$

$$1010_2 =$$

Conversión a Decimal

Binario → Decimal

$$\sum_{i=0}^n d_i \cdot 2^i$$

Ejemplo: 111101_2

$$1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 =$$

$$1 \times 32 + 1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 61$$

ejemplos simples:

$$101_2 = 5_{10}$$

$$1010_2 = 10_{10}$$

Conversión a Decimal

Binario → Decimal

$$\sum_{i=0}^n d_i \cdot 2^i$$

Ejemplo: 111101_2

$$1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 =$$

$$1 \times 32 + 1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 61$$

Hexa → Decimal

$$\sum_{i=0}^n d_i \cdot 16^i$$

Ejemplo: $3D_{16}$

$$3 \times 16^1 + 13 \times 16^0 =$$

$$3 \times 16 + 13 \times 1 = 61$$

Conversión a Decimal desde cualquier base

Base $b \rightarrow$ Decimal

$$\sum_{i=0}^n d_i \cdot b^i$$

Conversión Binario a...

... hexa

0 0 1 1 1 1 0 1
└───┘ └───┘
3 13

3D₁₆

... octal

1 1 1 1 0 1
└───┘ └───┘
7 5

75₈

Aritmética binaria

$$\begin{array}{rccccr} & & 1 & & & \\ & 1 & 0 & 0 & 1 & 9 \\ + & 0 & 1 & 0 & 1 & 5 \\ \hline & 1 & 1 & 1 & 0 & 14 \end{array}$$

$$\begin{array}{rccccr} & 0 & 1 & 1 & 1 & 7 \\ + & 0 & 1 & 0 & 0 & 4 \\ \hline & 1 & 0 & 1 & 1 & 11 \end{array}$$

Aritmética binaria

$$\begin{array}{r} \overset{1}{1} \ 0 \ 0 \ 1 \qquad 9 \\ + \ 0 \ 1 \ 0 \ 1 \qquad 5 \\ \hline 1 \ 1 \ 1 \ 0 \qquad 14 \end{array}$$

$$\begin{array}{r} \overset{1}{1} \ 1 \ 0 \ 0 \qquad 12 \\ + \ 0 \ 1 \ 0 \ 0 \qquad 4 \\ \hline \overset{1}{1} \ 0 \ 0 \ 0 \ 0 \qquad 16 \end{array}$$

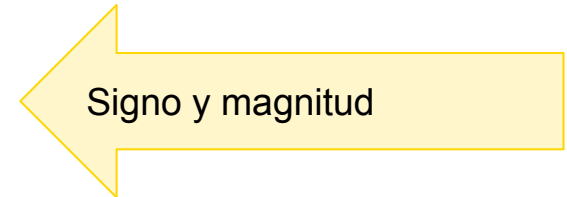
$$\begin{array}{r} \overset{1}{0} \ 1 \ 1 \ 1 \qquad 7 \\ + \ 0 \ 1 \ 0 \ 0 \qquad 4 \\ \hline 1 \ 0 \ 1 \ 1 \qquad 11 \end{array}$$

$$\begin{array}{r} \overset{1}{1} \ 1 \ 0 \ 0 \qquad 12 \\ + \ 1 \ 1 \ 1 \ 1 \qquad 15 \\ \hline \overset{1}{1} \ 1 \ 0 \ 1 \ 1 \qquad 27 \end{array}$$

Representación de números negativos

0 1 0 1 5

1 1 0 1 -5
└─┘ └─┘
Signo Magnitud



Representación de números negativos

0 1 0 1 5

1 1 0 1 -5
└─┘ └─┘
Signo Magnitud

Signo y magnitud

0 1 0 1 5

1 0 1 0 -5

Complemento a 1

Representación de números negativos

0 1 0 1 5

1 1 0 1 -5
└─┘ └─┘
Signo Magnitud

Signo y magnitud

0 1 0 1 5

1 0 1 0 -5

Complemento a 1

0 1 0 1 5

1 0 1 0
+ 0 0 0 1 +1
┌───┐
1 0 1 1 -5

Complemento a 2

Representación de números negativos

0 1 0 1 5

1 1 0 1 -5
└─┘ └─┘
Signo Magnitud

Signo y magnitud

0 1 0 1 5

1 0 1 0 -5

Complemento a 1

0 1 0 1 5

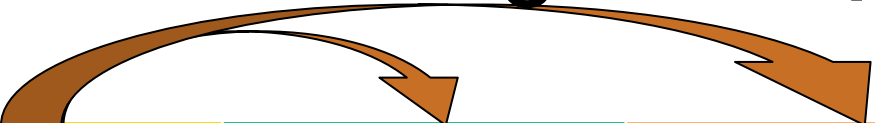
1 0 1 0
+ 0 0 0 1 +1
┌───┐
1 0 1 1 -5

Complemento a 2

1 0 1 1 -5

0 1 0 0
+ 0 0 0 1 +1
┌───┐
0 1 0 1 5

Interpretación de números negativos (4 bits)



Decimal	Binario	Complemento a 1	Complemento a 2
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	-7	-8
9	1001	-6	-7
10	1010	-5	-6
11	1011	-4	-5
12	1100	-3	-4
13	1101	-2	-3
14	1110	-1	-2
15	1111	0	-1

Aritmética binaria

Complemento a 2

$\begin{array}{r} 1\ 0\ 0\ 1 \\ + 0\ 1\ 0\ 1 \\ \hline 1\ 1\ 1\ 0 \end{array}$	$\begin{array}{r} 9\ -7 \\ 5\ 5 \\ 14\ -2 \end{array}$	$\begin{array}{r} 1\ 1\ 0\ 0 \\ + 0\ 1\ 0\ 0 \\ \hline 1\ 0\ 0\ 0\ 0 \end{array}$	$\begin{array}{r} 12 \\ 4 \\ 16 \end{array}$
--	--	---	--

$\begin{array}{r} 0\ 1\ 1\ 1 \\ + 0\ 1\ 0\ 0 \\ \hline 1\ 0\ 1\ 1 \end{array}$	$\begin{array}{r} 7 \\ 4 \\ 11 \end{array}$	$\begin{array}{r} 1\ 1\ 0\ 0 \\ + 1\ 1\ 1\ 1 \\ \hline 1\ 1\ 0\ 1\ 1 \end{array}$	$\begin{array}{r} 12 \\ 15 \\ 27 \end{array}$
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Aritmética binaria

Complemento a 2

$\begin{array}{r} \textcolor{blue}{1} \\ 1\ 0\ 0\ 1 \\ +\ 0\ 1\ 0\ 1 \\ \hline 1\ 1\ 1\ 0 \end{array}$	$\begin{array}{r} 9\ -7 \\ 5\ \textcolor{blue}{5} \\ 14\ -2 \end{array}$	$\begin{array}{r} \textcolor{blue}{1} \\ 1\ 1\ 0\ 0 \\ +\ 0\ 1\ 0\ 0 \\ \hline \textcolor{blue}{1}\ 0\ 0\ 0\ 0 \end{array}$	$\begin{array}{r} 12 \\ 4 \\ 16 \end{array}$
$\begin{array}{r} 0\ 1\ 1\ 1 \\ +\ 0\ 1\ 0\ 0 \\ \hline 1\ 0\ 1\ 1 \end{array}$	$\begin{array}{r} 4 \\ 11 \end{array}$	$\begin{array}{r} \textcolor{blue}{1} \\ 1\ 1\ 0\ 0 \\ +\ 1\ 1\ 1\ 1 \\ \hline \textcolor{blue}{1}\ 1\ 0\ 1\ 1 \end{array}$	$\begin{array}{r} 12 \\ 15 \\ 27 \end{array}$

$$\begin{array}{l} \sim (1\ 1\ 1\ 0) + 1 \\ (0\ 0\ 0\ 1) + 1 \\ (0\ 0\ 1\ 0) \end{array}$$

Aritmética binaria

Complemento a 2

$\begin{array}{r} 1 \\ 1\ 0\ 0\ 1 \\ +\ 0\ 1\ 0\ 1 \\ \hline 1\ 1\ 1\ 0 \end{array}$	$\begin{array}{r} 9\ -7 \\ 5\ 5 \\ 14\ -2 \end{array}$	$\begin{array}{r} 1 \\ 1\ 1\ 0\ 0 \\ +\ 0\ 1\ 0\ 0 \\ \hline 1\ 0\ 0\ 0\ 0 \end{array}$	$\begin{array}{r} 12 \\ 4 \\ 16 \end{array}$
--	--	---	--

$\begin{array}{r} 0\ 1\ 1\ 1 \\ +\ 0\ 1\ 0\ 0 \\ \hline 1\ 0\ 1\ 1 \end{array}$	$\begin{array}{r} 7\ 7 \\ 4\ 4 \\ 11\ ? \end{array}$	$\begin{array}{r} 1 \\ 1\ 1\ 0\ 0 \\ +\ 1\ 1\ 1\ 1 \\ \hline 1\ 1\ 0\ 1\ 1 \end{array}$	$\begin{array}{r} 12 \\ 15 \\ 27 \end{array}$
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Aritmética binaria

Complemento a 2

$$\begin{array}{r} \overset{1}{1} \ 0 \ 0 \ 1 \\ + \ 0 \ 1 \ 0 \ 1 \\ \hline 1 \ 1 \ 1 \ 0 \end{array}$$

$$\begin{array}{r} 9 \ -7 \\ 5 \ 5 \\ 14 \ -2 \end{array}$$

$$\begin{array}{r} \overset{1}{1} \ 1 \ 0 \ 0 \\ + \ 0 \ 1 \ 0 \ 0 \\ \hline \overset{1}{1} \ 0 \ 0 \ 0 \ 0 \end{array}$$

$$\begin{array}{r} 12 \ -4 \\ 4 \ 4 \\ 16 \ ? \end{array}$$

$$\begin{array}{r} \ 0 \ 1 \ 1 \ 1 \\ + \ 0 \ 1 \ 0 \ 0 \\ \hline 1 \ 0 \ 1 \ 1 \end{array}$$

$$\begin{array}{r} 7 \ 7 \\ 4 \ 4 \\ 11 \ ? \end{array}$$

$$\begin{array}{r} \overset{1}{1} \ 1 \ 0 \ 0 \\ + \ 1 \ 1 \ 1 \ 1 \\ \hline \overset{1}{1} \ 1 \ 0 \ 1 \ 1 \end{array}$$

$$\begin{array}{r} 12 \ -4 \\ 15 \ -1 \\ 27 \ ? \end{array}$$

$$\begin{array}{r} \ 0 \ 1 \ 0 \ 1 \\ - \ 0 \ 1 \ 0 \ 0 \\ \hline 0 \ 0 \ 0 \ 1 \end{array}$$

$$\begin{array}{r} 5 \\ 4 \\ 1 \end{array}$$

$$\begin{array}{r} \overset{1}{1} \ \overset{1}{1} \ 0 \ 0 \ 1 \\ - \ 1 \ 0 \ 1 \ 0 \\ \hline 1 \ 1 \ 1 \ 1 \end{array}$$

$$\begin{array}{r} 9 \\ 10 \\ ?? \end{array}$$

Aritmética binaria

Complemento a 2

$$\begin{array}{r}
 \textcolor{blue}{1} \\
 1\ 0\ 0\ 1 \\
 +\ 0\ 1\ 0\ 1 \\
 \hline
 1\ 1\ 1\ 0
 \end{array}$$

$$\begin{array}{r}
 9\ -7 \\
 5\ \textcolor{blue}{5} \\
 14\ -2
 \end{array}$$

$$\begin{array}{r}
 \textcolor{blue}{1} \\
 1\ 1\ 0\ 0 \\
 +\ 0\ 1\ 0\ 0 \\
 \hline
 \textcolor{blue}{1}\ 0\ 0\ 0\ 0
 \end{array}$$

$$\begin{array}{r}
 12\ -4 \\
 4\ \textcolor{blue}{4} \\
 16\ \textcolor{blue}{0}
 \end{array}$$

$$\begin{array}{r}
 0\ 1\ 1\ 1 \\
 +\ 0\ 1\ 0\ 0 \\
 \hline
 1\ 0\ 1\ 1
 \end{array}$$

$$\begin{array}{r}
 7\ \textcolor{blue}{7} \\
 4\ \textcolor{blue}{4} \\
 11\ \textcolor{red}{-5}
 \end{array}$$

$$\begin{array}{r}
 \textcolor{blue}{1} \\
 1\ 1\ 0\ 0 \\
 +\ 1\ 1\ 1\ 1 \\
 \hline
 \textcolor{blue}{1}\ 1\ 0\ 1\ 1
 \end{array}$$

$$\begin{array}{r}
 12\ -4 \\
 15\ -1 \\
 27\ -5
 \end{array}$$

$$\begin{array}{r}
 0\ 1\ 0\ 1 \\
 -\ 0\ 1\ 0\ 0 \\
 \hline
 0\ 0\ 0\ 1
 \end{array}$$

$$\begin{array}{r}
 5 \\
 4 \\
 1
 \end{array}$$

$$\begin{array}{r}
 \textcolor{blue}{1}\ \textcolor{blue}{1} \\
 \textcolor{blue}{1}\ 0\ 0\ 1 \\
 -\ 1\ 0\ 1\ 0 \\
 \hline
 1\ 1\ 1\ 1
 \end{array}$$

$$\begin{array}{r}
 9 \\
 10 \\
 \textcolor{red}{??}
 \end{array}$$

Aritmética binaria

Complemento a 2

¹ $\begin{array}{r} 1\ 0\ 0\ 1 \\ +\ 0\ 1\ 0\ 1 \\ \hline 1\ 1\ 1\ 0 \end{array}$	$\begin{array}{r} 9\ -7 \\ 5\ 5 \\ 14\ -2 \end{array}$	¹ $\begin{array}{r} 1\ 1\ 0\ 0 \\ +\ 0\ 1\ 0\ 0 \\ \hline 1\ 0\ 0\ 0\ 0 \end{array}$	$\begin{array}{r} 12\ -4 \\ 4\ 4 \\ 16\ 0 \end{array}$
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$\begin{array}{r} 0\ 1\ 1\ 1 \\ +\ 0\ 1\ 0\ 0 \\ \hline 1\ 0\ 1\ 1 \end{array}$	$\begin{array}{r} 7\ 7 \\ 4\ 4 \\ 11\ -5 \end{array}$	¹ $\begin{array}{r} 1\ 1\ 0\ 0 \\ +\ 1\ 1\ 1\ 1 \\ \hline 1\ 1\ 0\ 1\ 1 \end{array}$	$\begin{array}{r} 12\ -4 \\ 15\ -1 \\ 27\ -5 \end{array}$
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$\begin{array}{r} 0\ 1\ 0\ 1 \\ -\ 0\ 1\ 0\ 0 \\ \hline 0\ 0\ 0\ 1 \end{array}$	$\begin{array}{r} 5\ 5 \\ 4\ 4 \\ 1\ 1 \end{array}$	^{1\ 1} $\begin{array}{r} 1\ 0\ 0\ 1 \\ -\ 1\ 0\ 1\ 0 \\ \hline 1\ 1\ 1\ 1 \end{array}$	$\begin{array}{r} 9 \\ 10 \\ ?? \end{array}$
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Aritmética binaria

Complemento a 2

$$\begin{array}{r} \textcolor{blue}{1} \\ 1\ 0\ 0\ 1 \\ +\ 0\ 1\ 0\ 1 \\ \hline 1\ 1\ 1\ 0 \end{array}$$

$$\begin{array}{r} 9\ -7 \\ 5\ \textcolor{blue}{5} \\ 14\ -2 \end{array}$$

$$\begin{array}{r} \textcolor{blue}{1} \\ 1\ 1\ 0\ 0 \\ +\ 0\ 1\ 0\ 0 \\ \hline \textcolor{blue}{1}\ 0\ 0\ 0\ 0 \end{array}$$

$$\begin{array}{r} 12\ -4 \\ 4\ \textcolor{blue}{4} \\ 16\ \textcolor{blue}{0} \end{array}$$

$$\begin{array}{r} 0\ 1\ 1\ 1 \\ +\ 0\ 1\ 0\ 0 \\ \hline 1\ 0\ 1\ 1 \end{array}$$

$$\begin{array}{r} 7\ \textcolor{blue}{7} \\ 4\ \textcolor{blue}{4} \\ 11\ \textcolor{red}{-5} \end{array}$$

$$\begin{array}{r} \textcolor{blue}{1} \\ 1\ 1\ 0\ 0 \\ +\ 1\ 1\ 1\ 1 \\ \hline \textcolor{blue}{1}\ 1\ 0\ 1\ 1 \end{array}$$

$$\begin{array}{r} 12\ -4 \\ 15\ -1 \\ 27\ -5 \end{array}$$

$$\begin{array}{r} 0\ 1\ 0\ 1 \\ -\ 0\ 1\ 0\ 0 \\ \hline 0\ 0\ 0\ 1 \end{array}$$

$$\begin{array}{r} 5\ \textcolor{blue}{5} \\ 4\ \textcolor{blue}{4} \\ 1\ \textcolor{blue}{1} \end{array}$$

$$\begin{array}{r} \textcolor{blue}{1}\ \textcolor{blue}{1} \\ \textcolor{blue}{1}\ 0\ 0\ 1 \\ -\ 1\ 0\ 1\ 0 \\ \hline 1\ 1\ 1\ 1 \end{array}$$

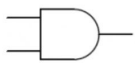
$$\begin{array}{r} 9\ -7 \\ 10\ -6 \\ \textcolor{red}{??}\ -1 \end{array}$$

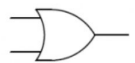


$$\begin{array}{r} 1\ 0\ 0\ 1 \\ +\ 0\ 1\ 1\ 0 \\ \hline 1\ 1\ 1\ 1 \end{array}$$

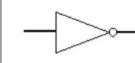
$$\begin{array}{r} 9\ -7 \\ 6\ \textcolor{blue}{6} \\ 15\ -1 \end{array}$$

Operaciones lógicas

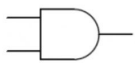
	AND
0 0	0
0 1	0
1 0	0
1 1	1

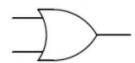
	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

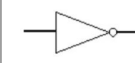
	NOT
0	1
1	0

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

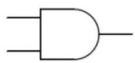
	NOT
0	1
1	0

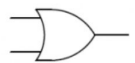
	NAND
0 0	1
0 1	1
1 0	1
1 1	0

	NOR
0 0	1
0 1	0
1 0	0
1 1	0

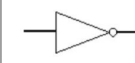
	XNOR
0 0	1
0 1	0
1 0	0
1 1	1

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

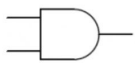
	NOT
0	1
1	0

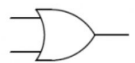
5 AND 3 = ?

5 OR 3 = ?

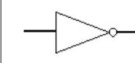
5 XOR 3 = ?

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

	NOT
0	1
1	0

$$5 \text{ AND } 3 = 1$$

$$5 \text{ OR } 3 = 7$$

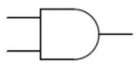
$$5 \text{ XOR } 3 = 6$$

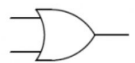
$$\begin{array}{r} 0101 \\ \text{AND } 0011 \\ \hline 0001 \end{array}$$

$$\begin{array}{r} 0101 \\ \text{OR } 0011 \\ \hline 0111 \end{array}$$

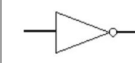
$$\begin{array}{r} 0101 \\ \text{XOR } 0011 \\ \hline 0110 \end{array}$$

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

	NOT
0	1
1	0

$$6 \text{ XOR } 3 = 5$$

$$\begin{array}{r} 0110 \\ \text{XOR } 0011 \\ \hline 0101 \end{array}$$

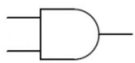
$$6 \text{ XOR } 5 = 3$$

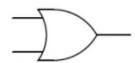
$$\begin{array}{r} 0110 \\ \text{XOR } 0101 \\ \hline 0011 \end{array}$$

$$5 \text{ XOR } 3 = 6$$

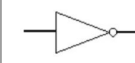
$$\begin{array}{r} 0101 \\ \text{XOR } 0011 \\ \hline 0110 \end{array}$$

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

	NOT
0	1
1	0

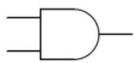
$$3D_{16} \text{ OR } 0F_{16} = 3F_{16}$$

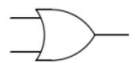
$$\begin{array}{r} 0011 \ 1101 \\ \text{OR } 0000 \ 1111 \\ \hline 0011 \ 1111 \end{array}$$

$$3D_{16} \text{ AND } 0F_{16} = 0D_{16}$$

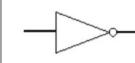
$$\begin{array}{r} 0011 \ 1101 \\ \text{AND } 0000 \ 1111 \\ \hline 0000 \ 1101 \end{array}$$

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

	NOT
0	1
1	0

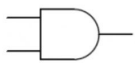
$$3D_{16} \text{ OR } 02_{16} = 3F_{16}$$

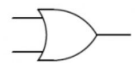
$$\begin{array}{r} 0011 \ 1101 \\ \text{OR } 0000 \ 0010 \\ \hline 0011 \ 1111 \end{array}$$

$$3D_{16} \text{ AND NOT}(02_{16}) = 3D_{16}$$

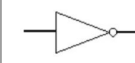
$$\begin{array}{r} 0011 \ 1101 \\ \text{AND } 1111 \ 1101 \\ \hline 0011 \ 1101 \end{array}$$

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

	NOT
0	1
1	0

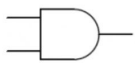
$$??_{16} \text{ OR } 02_{16} = ??_{16}$$

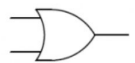
$$\begin{array}{r} \text{OR} \quad \begin{array}{cc} \text{????} & \text{????} \\ 0000 & 0010 \\ \hline \text{????} & \text{??1?} \end{array} \end{array}$$

$$??_{16} \text{ AND NOT}(02_{16}) = ??_{16}$$

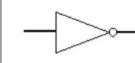
$$\begin{array}{r} \text{AND} \quad \begin{array}{cc} \text{????} & \text{????} \\ 1111 & 1101 \\ \hline \text{????} & \text{??0?} \end{array} \end{array}$$

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

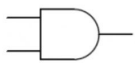
	XOR
0 0	0
0 1	1
1 0	1
1 1	0

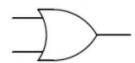
	NOT
0	1
1	0

$$\begin{array}{l}
 \text{??}_{16} \text{ AND } 02_{16} \\
 \begin{array}{l} \rightarrow = 00_{16} \\ \rightarrow \neq 00_{16} \end{array}
 \end{array}$$

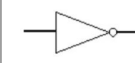
$$\begin{array}{r}
 \text{AND } \begin{array}{cc} \text{????} & \text{????} \\ 0000 & 0010 \end{array} \\
 \hline
 \begin{array}{cc} \text{????} & \text{????} \end{array}
 \end{array}$$

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

	NOT
0	1
1	0

'B' OR 20_{16} = ??

'a' AND DF_{16} = ??

Tabla ASCII

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

Tabla ASCII

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

Tabla ASCII

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	TX (transmission character)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	TX (transmission character)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	TX (transmission character)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	TX (transmission character)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	TX (transmission character)	40	28	048	A;	:	72	48	108	H	H	104	68	148	h	h
9	9	011	TX (transmission character)	41	29	049	B;	;	73	49	109	I	I	105	69	149	i	i
10	A	012	TX (transmission character)	42	2A	050	C;	<	74	4A	110	J	J	106	70	150	j	j
11	B	013	TX (transmission character)	43	2B	051	D;	=	75	4B	111	K	K	107	71	151	k	k
12	C	014	TX (transmission character)	44	2C	052	E;	>	76	4C	112	L	L	108	72	152	l	l
13	D	015	TX (transmission character)	45	2D	053	F;	?	77	4D	113	M	M	109	73	153	m	m
14	E	016	TX (transmission character)	46	2E	054	(	(	78	4E	114	N	N	110	74	154	n	n
15	F	017	TX (transmission character)	47	2F	055	)	)	79	4F	115	O	O	111	75	155	o	o
16	10	020	TX (transmission character)	48	30	056	*	*	80	50	116	P	P	112	76	156	p	p
17	11	021	DC1 (device control 1)	49	31	061	+	+	81	51	121	Q	Q	113	77	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	,	,	82	52	122	R	R	114	78	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	-	-	83	53	123	S	S	115	79	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	.	.	84	54	124	T	T	116	7A	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	/	/	85	55	125	U	U	117	7B	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	0	0	86	56	126	V	V	118	7C	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	1	1	87	57	127	W	W	119	7D	167	w	w
24	18	030	CAN (cancel)	56	38	070	2	2	88	58	130	X	X	120	7E	170	x	x
25	19	031	EM (end of medium)	57	39	071	3	3	89	59	131	Y	Y	121	7F	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	4	4	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	5	5	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	6	6	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	7	7	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	8	8	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	9	9	95	5F	137	_	_	127	7F	177		DEL

Operaciones lógicas

Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	
64	40	100	@	@	96	60	140	`	`	T
65	41	101	A	A	97	61	141	a	a	
66	42	102	B	B	98	62	142	b	b	
67	43	103	C	C	99	63	143	c	c	
68	44	104	D	D	100	64	144	d	d	
69	45	105	E	E	101	65	145	e	e	

'B' OR 20_{16} = ??

42_{16} OR 20_{16} =

'a' AND DF_{16} = ??

61_{16} AND DF_{16} =

Operaciones lógicas

Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	
64	40	100	@	@	96	60	140	`	`	T
65	41	101	A	A	97	61	141	a	a	
66	42	102	B	B	98	62	142	b	b	
67	43	103	C	C	99	63	143	c	c	
68	44	104	D	D	100	64	144	d	d	
69	45	105	E	E	101	65	145	e	e	

'B' OR 20_{16} = ??

42_{16} OR 20_{16} = 62_{16}

```

      0100 0010
OR   0010 0000
-----
      0110 0010

```

'a' AND DF_{16} = ??

61_{16} AND DF_{16} = 41_{16}

```

      0110 0001
AND   1101 1111
-----
      0100 0001

```


Operaciones lógicas

Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	OT
64	40	100	@	@	96	60	140	`	`	
65	41	101	A	A	97	61	141	a	a	
66	42	102	B	B	98	62	142	b	b	
67	43	103	C	C	99	63	143	c	c	
68	44	104	D	D	100	64	144	d	d	
69	45	105	E	E	101	65	145	e	e	

'B' OR 20_{16} = ??

42_{16} OR 20_{16} = 62_{16}

```

      0100 0010
OR   0010 0000
-----
      0110 0010
  
```

'a' AND DF_{16} = ??

61_{16} AND DF_{16} = 41_{16}

```

      0110 0001
AND   1101 1111
-----
      0100 0001
  
```

Operaciones lógicas

Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	
64	40	100	@	@	96	60	140	`	`	T
65	41	101	A	A	97	61	141	a	a	
66	42	102	B	B	98	62	142	b	b	
67	43	103	C	C	99	63	143	c	c	
68	44	104	D	D	100	64	144	d	d	
69	45	105	E	E	101	65	145	e	e	

'B' OR 20_{16} = 'b'

'a' AND DF_{16} = 'A'

42_{16} OR 20_{16} = 62_{16}

61_{16} AND DF_{16} = 41_{16}

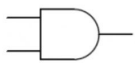
```

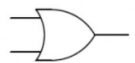
      0100 0010
OR   0010 0000
-----
      0110 0010
  
```

```

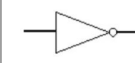
      0110 0001
AND  1101 1111
-----
      0100 0001
  
```

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

	NOT
0	1
1	0

'B' OR 20_{16} = 'b'

42_{16} OR 20_{16} = 62_{16}

$$\begin{array}{r} \text{OR} \quad \begin{array}{cc} 0100 & 0010 \\ 0010 & 0000 \\ \hline 0110 & 0010 \end{array} \end{array}$$

'a' AND NOT(20_{16}) = 'A'

61_{16} AND NOT(20_{16}) = 41_{16}

$$\begin{array}{r} \text{AND} \quad \begin{array}{cc} 0110 & 0001 \\ 1101 & 1111 \\ \hline 0100 & 0001 \end{array} \end{array}$$

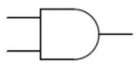
Tabla ASCII

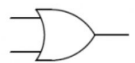
Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

Tabla ASCII

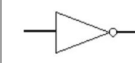
Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

Operaciones lógicas

	AND
0 0	0
0 1	0
1 0	0
1 1	1

	OR
0 0	0
0 1	1
1 0	1
1 1	1

	XOR
0 0	0
0 1	1
1 0	1
1 1	0

	NOT
0	1
1	0

'B' OR 'c' = 'b'

0100 0010
 OR 0010 0000

 0110 0010

42_{16} OR $20_{16} = 62_{16}$

'a' AND NOT('c') = 'A'

0110 0001
 AND 1101 1111

 0100 0001

61_{16} AND $DF_{16} = 41_{16}$

Operaciones de desplazamiento

Shift Left (SHL o <<)

$$0100_2 \text{ SHL } 1 = 1000_2$$

$$0101_2 \text{ SHL } 2 = 10100_2$$

$$3D_{16} \text{ SHL } 4 = D0_{16}$$

Shift right (SHR o >>)

$$0100_2 \text{ SHR } 1 = 0010_2$$

$$0101_2 \text{ SHR } 2 = 0001_2$$

$$3D_{16} \text{ SHR } 4 = 03_{16}$$

Operaciones de desplazamiento

Shift Left (SHL o <<)

$$0100_2 \text{ SHL } 1 = 1000_2$$

$$0101_2 \text{ SHL } 2 = 10100_2$$

$$3D_{16} \text{ SHL } 4 = D0_{16}$$

Shift right (SHR o >>)

$$0100_2 \text{ SHR } 1 = 0010_2$$

$$0101_2 \text{ SHR } 2 = 0001_2$$

$$3D_{16} \text{ SHR } 4 = 03_{16}$$

$$1100_2 \text{ SHR } 1 = 0110_2$$

Operaciones de desplazamiento

Shift Left (SHL o <<)

$$0100_2 \text{ SHL } 1 = 1000_2$$

$$0101_2 \text{ SHL } 2 = 10100_2$$

$$3D_{16} \text{ SHL } 4 = D0_{16}$$

Shift right (SHR o >>)

$$0100_2 \text{ SHR } 1 = 0010_2$$

$$0101_2 \text{ SHR } 2 = 0001_2$$

$$3D_{16} \text{ SHR } 4 = 03_{16}$$

$$1100_2 \text{ SHR } 1 = 0110_2$$

$$1100_2 \text{ SAR } 1 = 1110_2$$

Shift Arithmetic Right

Operaciones de desplazamiento

Shift Left (SHL o <<)

$$0100_2 \text{ SHL } 1 = 1000_2$$

$$0101_2 \text{ SHL } 2 = 10100_2$$

$$3D_{16} \text{ SHL } 4 = D0_{16}$$

Shift right (SHR o >>)

$$0100_2 \text{ SHR } 1 = 0010_2$$

$$0101_2 \text{ SHR } 2 = 0001_2$$

$$3D_{16} \text{ SHR } 4 = 03_{16}$$

$$1100_2 \text{ SHR } 1 = 0110_2$$

$$1100_2 \text{ SAR } 1 = 1110_2$$

$$0100_2 \text{ SAR } 1 = 0010_2$$

Shift Arithmetic Right

Ejemplos SH + Máscaras

Formato Fecha

2 Bytes 1 Byte 1 Byte

Año	Mes	Día
-----	-----	-----

Ejemplos SH + Máscaras

Formato Fecha

2 Bytes 1 Byte 1 Byte

Año	Mes	Día
-----	-----	-----

FECHA SHR 16_{10} = **AÑO**

Ejemplos SH + Máscaras

Formato Fecha

2 Bytes 1 Byte 1 Byte

Año	Mes	Día
-----	-----	-----

FECHA SHR 16_{10} = **AÑO**

FECHA AND FF_{16} = **DIA**

Ejemplos SH + Máscaras

Formato Fecha

2 Bytes 1 Byte 1 Byte

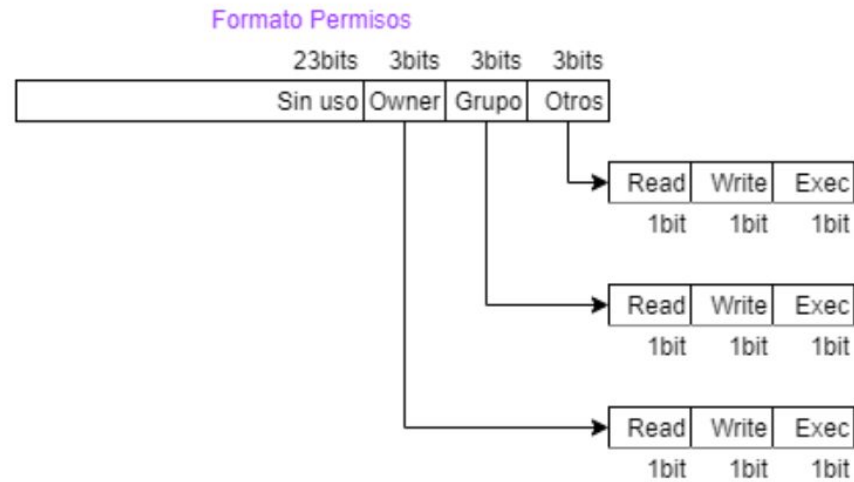
Año	Mes	Día
-----	-----	-----

FECHA SHR 16_{10} = **AÑO**

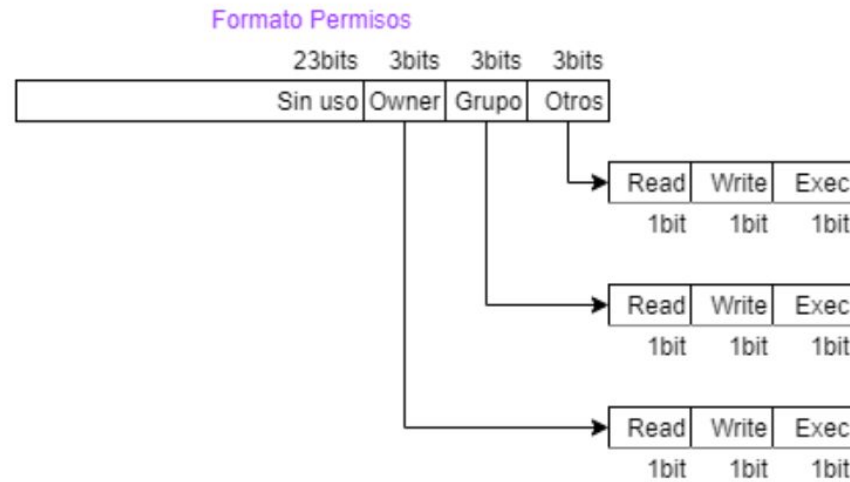
FECHA AND FF_{16} = **DIA**

(FECHA SHR 8_{10}) AND FF_{16} = **MES**

Ejemplos SH + Máscaras

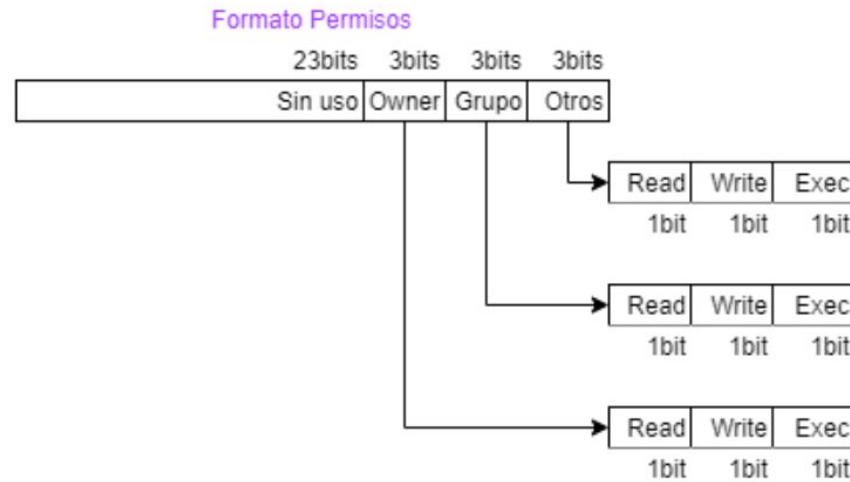


Ejemplos SH + Máscaras



PERMISOS = **PERMISOS** OR **777₈** //Asigna todos los permisos a todos.

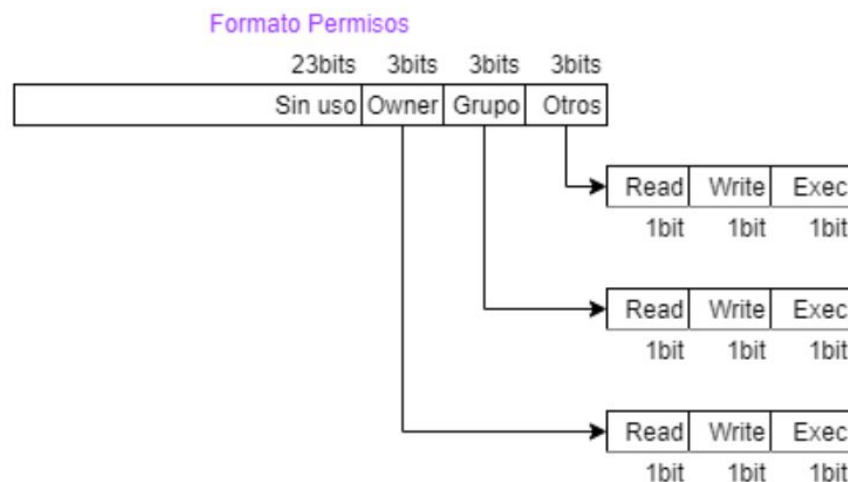
Ejemplos SH + Máscaras



PERMISOS = **PERMISOS** OR **777₈** //Asigna todos los permisos a todos.

PERMISOS = **PERMISOS** OR **700₈** //Asigna todos los permisos al owner y los demás quedan como estaban

Ejemplos SH + Máscaras



PERMISOS = **PERMISOS** OR **777₈** //Asigna todos los permisos a todos.

PERMISOS = **PERMISOS** OR **700₈** //Asigna todos los permisos al owner y los demás quedan como estaban

IF ((PERMISOS AND 010₈) != 0) //¿El grupo puede ejecutar?

Unidades de medida de la información

Tradicionalmente se utilizaba la nomenclatura decimal para las magnitudes binarias.

2^{10} bytes	1 kilobyte (kB)
2^{20} bytes	1 megabyte (MB)
2^{30} bytes	1 gigabyte (GB)
2^{40} bytes	1 terabyte (TB)
2^{50} bytes	1 petabyte (PB)
2^{60} bytes	1 exabyte (EB)
2^{70} bytes	1 zettabyte (ZB)
2^{80} bytes	1 yottabyte (YB)

Unidades de medida de la información

Múltiplos de bytes			
ISO/IEC 80000-13 (binario)		Sistema Internacional (decimal)	
Múltiplo (símbolo)	ISO/IEC	Múltiplo (símbolo)	SI
1 kibibyte (KiB)	2^{10} bytes	1 kilobyte (kB)	10^3 bytes
1 mebibyte (MiB)	2^{20} bytes	1 megabyte (MB)	10^6 bytes
1 gibibyte (GiB)	2^{30} bytes	1 gigabyte (GB)	10^9 bytes
1 tebibyte (TiB)	2^{40} bytes	1 terabyte (TB)	10^{12} bytes
1 pebibyte (PiB)	2^{50} bytes	1 petabyte (PB)	10^{15} bytes
1 exbibyte (EiB)	2^{60} bytes	1 exabyte (EB)	10^{18} bytes
1 zebibyte (ZiB)	2^{70} bytes	1 zettabyte (ZB)	10^{21} bytes
1 yobibyte (YiB)	2^{80} bytes	1 yottabyte (YB)	10^{24} bytes

Sistemas de Numeración,

Operaciones lógicas,
y otros...

Fundamentos de
Arquitectura de Computadoras

```
DECBINASCII
06501000001A
06601000010B
06701000011C
... ..
09701100001a
```